CALL LOG MERGER

Detailed Problem Statement

A phone application maintains call history records to help users review their previous calls. Sometimes, call records are stored in two separate call logs, and the application needs to combine them into one complete list. Each call record contains the caller's name and the duration of the call in minutes. The aim of this problem is to develop a C++ program that uses a singly linked list to store and merge these two call logs while preserving the order of the records.

The program should first create two separate singly linked lists. The user enters the number of records in the first call log and provides the name and duration for each call. The program creates a node for every record and inserts it at the end of the first list. The same process is then repeated for the second call log. Each node contains the caller's name, the call duration, and a pointer to the next node. This structure allows the program to connect records dynamically without requiring a fixed-size array.

After both call logs have been created, the program merges them by connecting the last node of the first linked list to the first node of the second linked list. This method keeps all records from the first log in their original order, followed by all records from the second log in their original order. If the first list is empty, the merged list can begin with the second list. The program then traverses the merged linked list from its first node to the end and displays each caller's name and call duration.

The input consists of the number of calls in the first log, the name and duration of each call, the number of calls in the second log, and the name and duration of each call in that log. For simple console input, caller names should be entered without spaces, and call durations should be entered as integer values in minutes. The output should display a heading such as “Merged Call Log” and list all records in the required order, showing each caller's name followed by the duration in minutes.

The main objective of this problem is to understand the creation, traversal, insertion, and merging of singly linked lists. It also demonstrates how linked lists can be used to combine dynamically stored records efficiently. The expected result is a single merged call log containing every record from both input lists, without changing the relative order of records within either original list. The programming language used is C++, and the primary data structure is a singly linked list.

Expected Outcome

The program should successfully accept two call logs, merge them into one linked list, and display the complete call history in order. For example, if the first log contains Amit's 5-minute

call and Neha's 8-minute call, while the second log contains Ravi's 3-minute call and Pooja's 6-minute call, the merged output should list Amit, Neha, Ravi, and Pooja in that sequence with their corresponding durations.